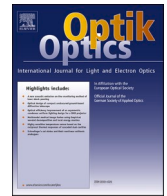




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Original research article

Robust hybrid modulated massive MIMO-FSOC frame for reliable remote healthcare monitoring in atmospheric environment

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ABSTRACT

High – capacity and reliable wireless communication is essential for Internet of Things (IoT) - enabled healthcare monitoring systems operating in license-free spectrum environments. Free –Space Optical Communication (FSOC) provides high bandwidth and immunity to electromagnetic interference, but its performance is significantly affected by atmospheric turbulence. This work proposes a hybrid-modulated spatial diversity-based MIMO-FSOC multi - aperture architecture for robust transmission of medical data under dynamically varying atmospheric conditions. The system integrates On-Off Keying (OOK) and Offset Quadrature Phase Shift Keying (OQPSK) to enhance spectral efficiency and transmission reliability. The system supports 110 Gbps data transmission and is evaluated over a Gamma-Gamma (GG) turbulence channel under haze, rain, snow, and fog conditions with and without ambient noise. Performance evaluation in terms of BER, Q-factor, and received power demonstrates improved reliability and extended communication range compared to conventional systems. The simulation results show that the achievable link distances under haze conditions are 10500 m without ambient loss and 9550 m with ambient loss. Under rain conditions, link distances of 7100 m without ambient loss and 6550 m with ambient loss are achieved. For snow conditions, the supported ranges are 5700 m and 4950 m without and with ambient loss, respectively. In fog environments, link distances of 3150 m without ambient loss and 2,900 m with ambient loss are obtained. These results demonstrate enhanced reliability and extended link range over conventional FSOC systems, confirming the effectiveness of the proposed multi-aperture optical MIMO- FSOC framework for medical data transmission under adverse atmospheric conditions.

1. Introduction

In modern era of Artificial Intelligence (AI), the rapid growth of data-driven applications has significantly increased the demand for IoT connectivity and efficient data exchange mechanisms. FSOC provides a high-data-rate, point-to-point optical wireless solution with relatively simple deployment over short to moderate link distances. Despite its inherent advantages, including large bandwidth availability, low power consumption, reduced electromagnetic interference, unlicensed spectrum, and enhanced physical-layer security, the performance of FSOC systems is strongly affected by atmospheric impairments. Crucial limiting factors include atmospheric turbulence, rain, haze, fog, snow, and background radiation from solar and sky irradiance, all of which degrade system performance in

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